

PSDA Assignment 3

1. a) Point estimate of the population mean, μ .

$$\hat{p} = \frac{345 + 420 + 390 + 410 + 380 + 400 + 430 + 360 + 375 + 390 + 405 + 415 + 400 + 385 + 395}{15}$$

$$= \frac{5900}{15}$$

$$= 393.333$$

- b) 95% $\alpha = 1.96$

$$\bar{x} \pm (t_{\text{critical value}}) \left(\frac{s}{\sqrt{n}} \right)$$

$$s = \sqrt{\frac{(345-393.333)^2 + (420-393.333)^2 + (390-393.333)^2 + (410-393.333)^2 + (380-393.333)^2 + (400-393.333)^2 + (430-393.333)^2 + (360-393.333)^2 + (375-393.333)^2 + (390-393.333)^2 + (405-393.333)^2 + (415-393.333)^2 + (400-393.333)^2 + (385-393.333)^2 + (395-393.333)^2}{15-1}}$$

$$= \sqrt{\frac{7083.333}{14}}$$

$$= 22.493$$

$$\bar{x} \pm (t_{\text{critical value}}) \left(\frac{s}{\sqrt{n}} \right)$$

$$393.333 \pm (2.145) \left(\frac{22.493}{\sqrt{15}} \right) = 393.333 \pm (12.457)$$

$$= (380.876, 405.790)$$

$\therefore$ We are 95% confident that the true population mean monthly expenditure lies in interval (380.876, 405.790)

2. a) $H_0: \mu = 5$

$$H_1: \mu < 5$$

(c) $\alpha = 0.01$

critical value = -2.32

b) $z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$

$$z = \frac{4.6 - 5}{\frac{1.2}{\sqrt{500}}}$$

$$= -7.454$$

(d) $-7.454 < -2.32$, reject null hypothesis

e) There is sufficient evidence that the average waiting time for customer in bank's queue is less than 5 minutes by using significant level 0.01

3. a) $H_0 : \mu = 160$
 $H_1 : \mu > 160$

c) $\alpha = 0.05$
 $(n-1) = (25-1)$
 $= 24$

b) $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$

$t_{0.05, 24} = 1.711$

$t = \frac{183 - 160}{\frac{12}{\sqrt{25}}}$
 $= 9.583$

d) Since $9.583 > 1.711$, we reject H_0 .

e) There is sufficient evidence that the mean score for the students from this university is greater than 160.

4. a) $H_0 : \mu = 50$
 $H_1 : \mu > 50$

c) $\alpha = 0.01$
 Z critical value = 2.33

b) $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

$z = \frac{51 - 50}{\frac{3.6}{\sqrt{38}}}$
 $= 1.712$

d) Since $1.712 < 2.33$, failed to reject null hypothesis.

e) There is sufficient evidence that support the population mean is 50 cm.

5. a) $H_0 : \sigma = 7460$
 $H_1 : \sigma > 7460$

c) $\alpha = 0.01$
 $\chi^2_{0.01, 51} = 76.154$

b) $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$

$= \frac{(51-1) 7871^2}{(7460)^2}$
 $= 55.661$

d) Since $55.661 < 76.154$, failed to reject H_0 .

e) There is sufficient evidence that support the males have a standard deviation that is same to the standard deviation of 7460 words for females.

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6. a) Female :

$$\begin{aligned}\bar{x}_1 &= \frac{4.3 + 4.3 + 4.4 + 4.0 + 3.4 + 4.7 + 2.9 + 4.0 + 4.3 + 3.4 + 3.4 + 3.3}{12} \\ &= \frac{46.4}{12} \\ &= 3.867\end{aligned}$$

$$\begin{aligned}S_1 &= \sqrt{\frac{(4.3-3.867)^2 + (4.3-3.867)^2 + (4.4-3.867)^2 + (4.0-3.867)^2 + (3.4-3.867)^2 + (4.7-3.867)^2 + (2.9-3.867)^2 + (4.0-3.867)^2 + (4.3-3.867)^2 + (3.4-3.867)^2 + (3.4-3.867)^2 + (3.3-3.867)^2}{12-1}} \\ &= 0.563\end{aligned}$$

Male :

$$\begin{aligned}\bar{x}_2 &= \frac{4.5 + 3.7 + 4.2 + 3.9 + 3.1 + 4.0 + 3.8 + 3.4 + 4.5 + 3.8 + 4.3 + 4.4 + 4.1 + 4.2 + 4.0 + 4.4}{16} \\ &= \frac{64.3}{16} \\ &= 4.019\end{aligned}$$

$$\begin{aligned}S_2 &= \sqrt{\frac{(4.5-4.019)^2 + (3.7-4.019)^2 + (4.2-4.019)^2 + (3.9-4.019)^2 + (3.1-4.019)^2 + (4.0-4.019)^2 + (3.8-4.019)^2 + (3.4-4.019)^2 + (4.5-4.019)^2 + (3.8-4.019)^2 + (4.3-4.019)^2 + (4.4-4.019)^2 + (4.1-4.019)^2 + (4.2-4.019)^2 + (4.0-4.019)^2 + (4.4-4.019)^2}{16-1}} \\ &= 0.395\end{aligned}$$

$$\begin{aligned}b) H_0 : \mu_1 - \mu_2 &= \Delta_0 = 0 \\ H_1 : \mu_1 &\neq \mu_2\end{aligned}$$

assume σ equal.

$$\alpha = 0.05$$

$$df, v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2-1}}$$

$$v = \frac{\left(\frac{0.563^2}{12} + \frac{0.395^2}{16}\right)^2}{\frac{(0.563^2)^2}{12-1} + \frac{(0.395^2)^2}{16-1}}$$

$$= 18.747$$

$$\approx 19$$

Reject H_0 if $t_0 > t_{0.025, 19} = 2.093$

or

if $t_0 < -t_{0.025, 19} = -2.093$

why?

$$\begin{aligned}
 S_p^2 &= \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \\
 &= \frac{(12 - 1)(0.563)^2 + (16 - 1)(0.395)^2}{12 + 16 - 2} \\
 &= 0.2241166923
 \end{aligned}$$

$$\begin{aligned}
 S_p &= \sqrt{0.2241166923} \\
 &= 0.473
 \end{aligned}$$

$$\begin{aligned}
 T_o &= \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \\
 &= \frac{3.867 - 4.019 - 0}{0.473 \sqrt{\frac{1}{12} + \frac{1}{16}}} \\
 &= -0.841
 \end{aligned}$$

c) Conclusion : Since $-2.093 < t_o = -0.841 < 2.093$, fail to reject H_0 . At 0.05 level significance, we do not have strong evidence to conclude that female professors in a mean evaluation scores of course evaluation taught that differs from the mean evaluation scores of course evaluation taught by male professors.

$$d) H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_1 : \sigma_1^2 > \sigma_2^2$$

$$\alpha = 0.01$$

$$v_1 = 12 - 1$$

$$v_2 = 16 - 1$$

$$= 11 \text{ (numerator)}$$

$$= 15 \text{ (denominator)}$$

$$\begin{aligned}
 F_{0.01, 11, 15} &= \frac{3.80 + 3.67}{2} \\
 &= 3.735
 \end{aligned}$$

$$\begin{aligned}
 F &= \frac{S_1^2}{S_2^2} \\
 &= \frac{0.563^2}{0.395^2} \\
 &= 2.032
 \end{aligned}$$

Conclusion : Since $2.032 < 3.735$, fail to reject H_0 . There is insufficient evidence to conclude that the variance of the course evaluation scores for female professors is greater than the male professors.

7. a)

Height(m) (x_1)	Height(d) (x_2)	$D = x_1 - x_2$	$D^2 = (x_1 - x_2)^2$
68.0	68.5	-0.5	0.25
60.0	60.0	0.0	0.00
61.0	63.5	-2.5	6.25
63.5	67.5	-4.0	16.00
69.0	68.0	1.0	1.00
64.0	65.5	-1.5	2.25
69.0	69.0	0.0	0.00
64.0	68.0	-4.0	16.00
63.5	64.5	-1.0	1.00
66.0	63.0	3.0	9.00
		$\Sigma D = -9.5$	$\Sigma D^2 = 51.75$

$$H_0: \mu_D = 0$$

$$H_1: \mu_D \neq 0$$

$$\text{degree freedom, } df = n - 1$$

$$= 10 - 1$$

$$= 9$$

$$t < -2.262 \text{ or } t > 2.262$$

$$b) \bar{D} = \frac{\Sigma D}{n} = \frac{-9.5}{10} = -0.95$$

$$s_D = \sqrt{\frac{\Sigma D^2 - \frac{(\Sigma D)^2}{n}}{n - 1}} = \sqrt{\frac{51.75 - \frac{(-9.5)^2}{10}}{10 - 1}} = 2.179$$

$$c) t = \frac{\bar{D} - \mu_D}{\frac{s_D}{\sqrt{n}}} = \frac{-0.95 - 0}{\frac{2.179}{\sqrt{10}}} = -1.379$$

$\therefore$ Since $t < -2.262$, $-1.379 > 2.262$, fail to reject H_0 .

At 0.05 level significance, we do not have strong evidence to conclude that there is no difference in heights between mothers and their first daughters.

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		Total cell i $\times$ Total cell j		total	
8. a)	Cell, ij	Observed Count, o_{ij}	Expected Count, e_{ij}	$(o_{ij} - e_{ij})^2 / e_{ij}$	
	1, 1	90	93.023	0.098	
	1, 2	80	79.070	0.011	
	1, 3	30	27.907	0.157	
	2, 1	70	69.761	0.001	
	2, 2	60	59.302	0.008	
	2, 3	20	20.930	0.041	
	3, 1	40	37.509	0.209	
	3, 2	30	31.628	0.084	
	3, 3	10	11.163	0.121	
			$\sum \chi^2$	0.730	

b) H_0 : The relationship between customer satisfaction levels and preferable method of communication is independent.

H_1 : The relationship between customer satisfaction levels and preferable method of communication is dependent.

Test Statistic : $\chi^2 = 0.730$

$$\alpha = 0.05$$

$$d.f. = (r-1)(c-1)$$

$$= (3-1)(3-1)$$

$$= 4$$

$$\text{Critical value : } \chi^2_{k=4, \alpha=0.05} = 9.488$$

$\therefore$ Since $0.730 < 9.488$, fail to reject H_0 . There is no strong evidence to conclude that the relationship between customer satisfaction levels and preferable method of communication is independent at 0.05 significance level.